

Graph Traversals (BFS & DFS) Data Structures

> **Definition:** Graph Traversal is the process of visiting all vertices reachable from a starting vertex in a graph. The two fundamental search algorithms are **Breadth-First Search (BFS)** and **Depth-First Search (DFS)**.

1. Detailed Technical Explanation

```

### 1. Breadth-First Search (BFS)

- **Strategy:** Explores neighbor vertices level-by-level using a **FIFO Queue**.

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### 2. Depth-First Search (DFS)

- **Strategy:** Explores as deep as possible along each branch before **backtracking**, using **Recursion** or a **Stack**.

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## 2. Complete Executable C Program for BFS & DFS

```
```c
```

```
#define MAX 10
```

```
int adj[MAX][MAX];
```

```
// 1. Breadth-First Search (BFS)
```

```
visitedBFS[start] = true;
```

```
printf("BFS Traversal: ");
```

```
for (int v = 0; v < V; v++) {
```

```
// 2. Depth-First Search (DFS)
```

```
for (int v = 0; v < V; v++) {
```

```
int main() {
```

```
BFS(0);
```

```
for (int i = 0; i < V; i++) visited[i] = false;
```

```
return 0;
```

```
bool vi
```

```
void B
```

```
queue
```

```
while (
```

```
if (adj[
```

```
void D
```

```
if (adj[
```

```
// Cons
```

```
printf("
```

```
}
```

3. Comparison of BFS and DFS

| Feature | Breadth-First Search (BFS) | Depth-First Search (DFS) |

| :--- | :--- |

4. Quick Recall Flow

'''

BFS ->